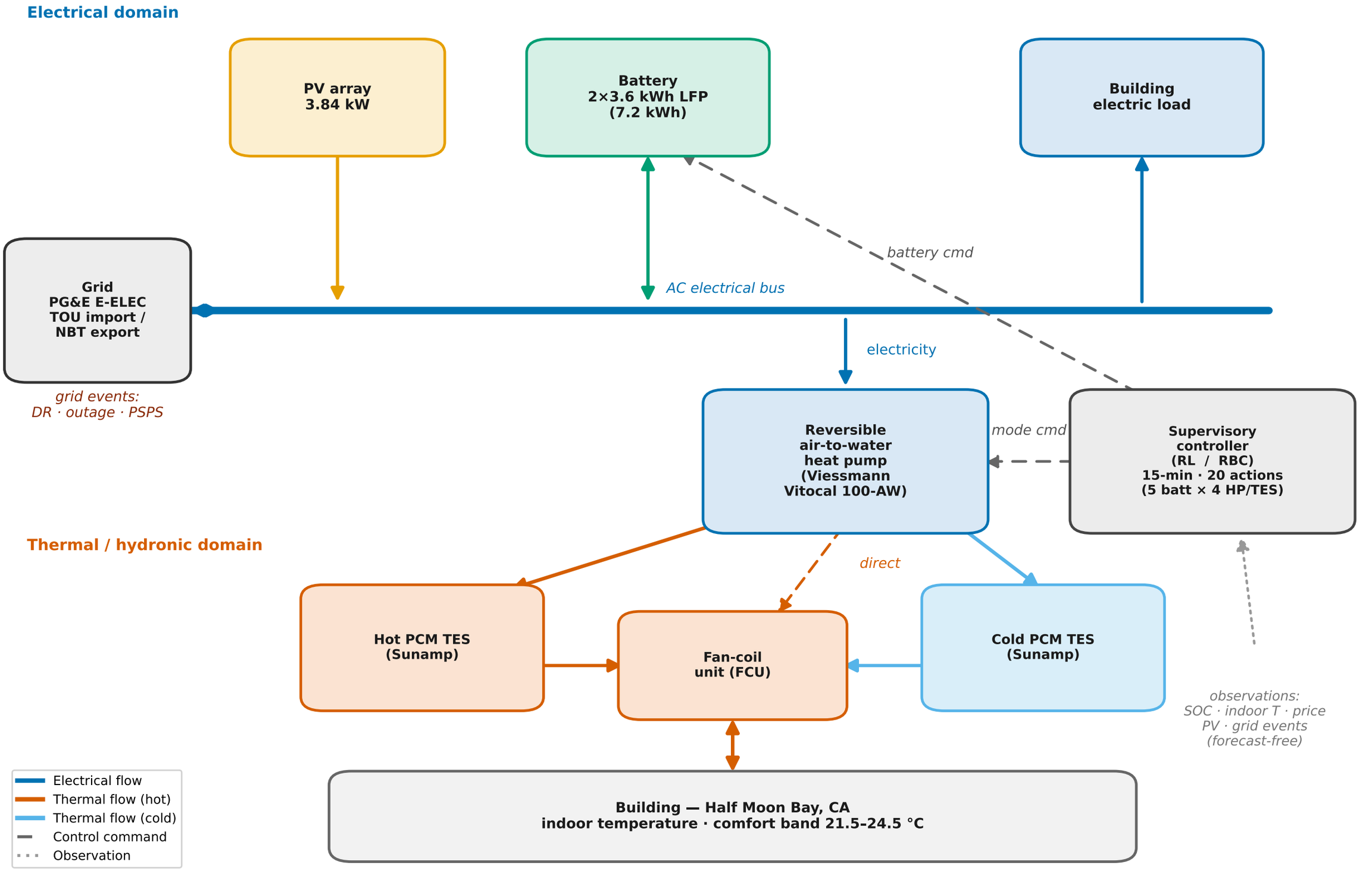
Electrical domain
PV array
3.84 kW
Battery
2×3.6 kWh LFP
(7.2 kWh)
Building
electric load
Grid
PG&E E-ELEC
TOU import /
NBT export
battery cmd
AC electrical bus
electricity
grid events:
DR · outage · PSPS
Reversible
air-to-water
heat pump
(Viessmann
Vitocal 100-AW)
mode cmd
Supervisory
controller
(RL / RBC)
15-min · 20 actions
(5 batt × 4 HP/TES)
Thermal / hydronic domain
direct
Hot PCM TES
(Sunamp)
Fan-coil
unit (FCU)
Cold PCM TES
(Sunamp)
observations:
SOC · indoor T · price
PV · grid events
(forecast-free)
Building — Half Moon Bay, CA
indoor temperature · comfort band 21.5-24.5 °C
Electrical flow
Thermal flow (hot)
Thermal flow (cold)
Control command
Observation